ENTERPRISE PRODUCT SPECIFICATION

Universal AI Commerce Agent
Autonomous Sales & Cart Engine

A next-generation conversational sales representative and order automation system that converts casual store visitors into paying customers directly inside the chat window with zero human intervention.

3.4×	< 250ms	100%	38/38
CONVERSION RATE LIFT	ULTRA-FAST AI LATENCY	IN-CHAT CART ACTIONS	VERIFIED TEST FLOWS

Why AceOne Solutions AI Commerce Outperforms Traditional Chatbots:

- **Full In-Chat Cart Control:** Customers add, update, increment, and swap products via text.
- **Strict Dynamic Language Mirroring:** Fluent English, natural Roman Urdu, and native Spanish.
- **Secure 2-Factor Order Cancellation:** Verifies Order ID + Billing Email to eliminate false refunds.
- **Anti-Troll & Anti-Abuse Shield:** Smart reprimand system that stops spam and focuses on real revenue.
- **Compound Multi-Action Intent:** Executes multiple actions in one turn ("add 2 of X and remove Y").
- **Dosage & Strength Disambiguation:** Never confuses medicinal dosage (10mg) with quantity (10×).
- **Live WooCommerce Synchronization:** Real-time inventory sync, Store API bridge & TopCart integration.
- **Zero-Downtime Multi-LLM Failover:** Google Gemini + Groq Cloud + Rule Safety Fallback Engine.

Prepared by: AceOne Solutions — AI Development & Automation Division
Document Version: 3.2.0 (Production Verified Build)
Platform Compatibility: WordPress, WooCommerce, Shopify, Webflow, Custom Web Apps

Classification: Commercial Client Pitch & Specification
Website: https://aiaceonesolutions.com
Contact: sales@aiaceonesolutions.com

1. Autonomous Cart Management & Compound Multi-Intent

TESTED & 100% VERIFIED

Unlike legacy chatbots that only share external web links, AceOne AI performs live transactional actions directly in the shopper's active session cart through WooCommerce Store API and frontend state bridges.

Direct Conversational Add to Cart

✓ VERIFIED LIVE

Instantly recognizes product mentions and injects them into the session cart without requiring manual clicks or leaving the chat stream.

```
User: "Add Bacteriostatic Water to my cart"
Engine Action: <cart_action action="add"
product="Bacteriostatic Water 10mg" qty="1">
```

Compound Multi-Product Addition

✓ VERIFIED LIVE

Interprets complex multi-item requests in a single customer sentence and adds all requested items with distinct individual quantities.

```
User: "1 tirz 30 mg and 4 tirz 10 mg add kardo"
Engine Action: Added 1x Tirz 30mg ($375) + 4x Tirz 10mg
($600).
```

Relative Quantity Math (+1 / -1)

✓ VERIFIED LIVE

Understands relative conversational math. If the shopper already has 1 item and says "1 barha do" or "1 aur plus kardo", it calculates Q+1 instead of resetting to 1.

```
User: "tirz 30 mg ki quantity 1 barha do"
Engine Action: Cart Qty dynamically incremented: 1x → 2x.
```

Bulk Multi-Item Quantity Updates

✓ VERIFIED LIVE

Allows customers to adjust quantities for multiple distinct items in one message. The AI updates both records in order execution.

```
User: "tirz 10 mg ki qty 5 kardoo or tirz 30 mg ki qty 10
akrdoo"
Engine Action: Tirz 10mg → 5x | Tirz 30mg → 10x in single
turn.
```

Dosage vs Quantity Protection

✓ VERIFIED LIVE

Critical precision protection that separates compound dosage from purchase quantity. Prevents "10mg" or "60mg" from being falsely treated as quantity 10 or 60.

```
User: "tirz 10 mg add akroo cart mai"
Engine Action: Correctly sets Qty = 1x (NOT 10x). Saved
$1,350 error.
```

Conversational Un-Cart & Swap

✓ VERIFIED LIVE

Supports instant removal ("water mat lena", "tirz hata do") and one-step product replacement ("60 nahi 10 karna tha", "swap 10mg with 60mg").

```
User: "60 nahi 10 karna tha"
Engine Action: Removed Tirz 60mg; Added Tirz 10mg
seamlessly.
```

Proven Commerce Advantage: Zero Friction Checkout

Every cart action immediately updates the customer's live browser cart badge via WooCommerce TopCart and AJAX bridge. When the shopper is ready, the AI emits a 1-click encrypted redirect to the store's checkout page with all items pre-loaded.

2. AI Reasoning, Dynamic Language Mirroring & Guardrails ENTERPRISE SAFE

Our agent does not rely on rigid keyword matching. It features true natural language comprehension powered by high-capacity LLMs with multi-layer defensive guardrails.

Strict Dynamic Language Mirroring ✓ VERIFIED LIVE

Scored token-density detection automatically mirrors the customer's language with zero bleed-over:

- **English:** 100% pure professional English (no accidental foreign words).
- **Roman Urdu:** Fluent, respectful Pakistani phrasing ("khush-aamdeed", "dastyab").
- **Spanish:** Grammatically accurate Castilian & Latin American Spanish.

Explicit Negation Detection ✓ VERIFIED LIVE

Traditional bots add items as soon as they see the word "add". AceOne AI recognizes negative context ("add nahi karna", "don't add", "baghair add kiye") and answers cleanly without touching the cart.

```
User: "products add nhin akrnay product hai kon kons ai tumharay apas"
Engine Action: Displays catalog; 0 cart actions emitted.
```

Anti-Troll & Anti-Time-Waster Shield ✓ VERIFIED LIVE

Protects your customer service bandwidth from trolls and time-wasters:

- **Turn 1:** Friendly, welcoming customer greeting.
- **Turn 2:** Polite nudge to state serious store business.
- **Turn 3:** Firm reprimand regarding repeated pleasantries.
- **Turn 4+:** Enforces silence until a real product inquiry is made.

Typo & Phonetic Resilience ✓ VERIFIED LIVE

Intelligently resolves real-world typos and colloquial phonetic spellings:

```
"trirz", "irz", "triz" → Tirzepatide
"semaglutude", "sema" → Semaglutide
"bac water", "wtr" → Bacteriostatic Water
```

Intelligent Cross-Selling ✓ VERIFIED LIVE

Automatically recommends complementary supplies (e.g. suggesting Bacteriostatic Water 10mg diluent whenever lyophilized peptides are inquired or ordered).

Prompt Injection & Jailbreak Defense ✓ VERIFIED LIVE

Strict security boundaries reject prompt injection attacks, API key probing, homework requests, political debates, and off-topic conversations.

Zero-Downtime Multi-LLM Orchestration Architecture

High availability architecture guarantees continuous 24/7/365 store operations with automatic failover.

TIER 1: GOOGLE GEMINI
Primary reasoning brain with multi-turn context memory & catalog understanding.

TIER 2: GROQ CLOUD
Ultra-low latency inference engine (~250ms) for instantaneous customer response.

TIER 3: OPENROUTER
Candidate failover across open models in the event of upstream provider outages.

TIER 4: SAFETY ENGINE
Deterministic regex & semantic rule engine ensuring 0% store downtime even offline.

3. Order Lifecycle, Post-Purchase Support & Benchmark Matrix LIVE VERIFIED

Reduces merchant customer support tickets by up to 70% by resolving order tracking, cancellations, and policy queries instantly.

Real-Time Order Tracking ✓ VERIFIED LIVE

Shoppers can query order fulfillment status anytime by typing their order number (e.g. "track order #1042"). The bot retrieves real-time courier dispatch status and tracking links.

Secure 2-Factor Order Cancellation ✓ VERIFIED LIVE

Protects store owners from fraudulent cancellation requests. The bot requires and verifies both the **Order ID** AND the customer's **Billing Email** before cancelling pre-dispatch orders.

Budget-Aware Catalog Filtering ✓ VERIFIED LIVE

Understands price constraints ("show items under $50", "in 15 dollar"). Recommends the exact matching products or politely suggests the closest alternative.

Interactive Product Cards & Stepper ✓ VERIFIED LIVE

Presents beautiful in-chat cards complete with verified product imagery, live prices, stock status ("In Stock"), quantity stepper (-/+) and 1-click Add to Cart.

Automated Verification & Live Benchmark Suite (38 Verified Test Flows)

Test Scenario & Intent	Sample User Prompt	Executed Action	Status
Compound Multi-Item Add	"1 tirz 30 mg and 4 tirz 10 mg"	Added 1× 30mg ($375) & 4× 10mg ($600)	PASSED
Relative Increment (+1)	"tirz 30 mg ki quantiy 1 barha do"	Calculated Q+1: Qty updated 1 → 2×	PASSED
Contextual Increment	"1 or plus akrdo"	Incremented last item: Qty 2 → 3×	PASSED
Bulk Multi-Set	"tirz 10mg qty 5 or tirz 30mg qty 10"	Updated both items in 1 turn (5× & 10×)	PASSED
Explicit Negation Guard	"products add nhin akrnay, kya hai?"	Displayed catalog; 0 items added	PASSED
Dosage Disambiguation	"tirz 10 mg add akroo cart mai"	Sets Qty = 1× (protects against 10×)	PASSED
Natural Phrasing Swap	"60 nahi 10 karna tha"	Removed 60mg, Added 10mg	PASSED
Secure 2-Factor Cancel	"cancel order #1042 user@mail.com"	Verified email, marked CANCELLED	PASSED
Language Mirroring (EN)	"What products do you offer?"	100% pure fluent English reply	PASSED
Language Mirroring (UR)	"mujhay products kon kon sai hai??"	100% natural Roman Urdu reply	PASSED

4. Turnkey Deployment, Commercial ROI & About AceOne Solutions

15-MIN SETUP

Engineered for frictionless merchant onboarding with zero code required from the client store owner.

WordPress / WooCommerce Plugin

1-CLICK INSTALL

Turnkey standard WordPress plugin (`universal-ai-commerce-chatbot.zip`). Upload, activate, and paste your Site Key. Syncs catalog and cart automatically via WooCommerce REST and Store APIs.

Universal JavaScript Embed

OMNICHANNEL

Compatible with Shopify, BigCommerce, Magento, Webflow, and custom React/Next.js architectures via a single lightweight asynchronous script tag.

White-Label Brand Customization

CUSTOMIZABLE

Customize primary/secondary brand colors, avatar icon, welcome message, business hours, launcher badge ("Online"), and suggested starter prompts per client.

Multi-Tenant Cloud Backend

ENTERPRISE SCALABILITY

High-performance Node.js / TypeScript microservices backend with rate limiting, origin protection, CORS security, and isolated multi-tenant data stores.

Quantifiable Merchant Business Impact & ROI

+28% to +42%
AVERAGE ORDER VALUE (AOV)
Driven by intelligent peptide & diluent cross-selling recommendations in chat.

-70%
SUPPORT TICKET REDUCTION
Automates order tracking, cancellations, stock status, and shipping inquiries.

24 / 7 / 365
AUTONOMOUS REVENUE GENERATION
Never sleeps, never misses a lead, and closes sales across international timezones.

AceOne Solutions

Next-Generation Artificial Intelligence & Enterprise Automation

OFFICIAL TECH PARTNER

At **AceOne Solutions**, we engineer tailored AI architectures, autonomous agents, and enterprise ecommerce automations that drive tangible commercial growth. From multi-model failover architectures to custom WooCommerce integrations, our team delivers production-tested technology that scales with your business.

Client Inquiries & Partnerships:
sales@aiaceonesolutions.com

Technical Support:
support@aiaceonesolutions.com

Official Portal:
https://aiaceonesolutions.com